

AI-Powered Causal Crime Reconstruction Engine (CCRE)

Advanced Cyber Incident Reconstruction and Graph Analysis System

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Problem Statement

- Modern cyber attacks are multi-stage, stealthy, and highly distributed.
- Security logs are massive, noisy, and often disconnected across systems.
- Analysts must manually correlate thousands of SIEM alerts to reconstruct incidents.
- Traditional workflows are slow, error-prone, and heavily analyst-dependent.
- Identifying root cause and attack progression remains difficult in real time.

Why This System is Needed

- Cyber attacks are increasing in scale, complexity, and sophistication.
- Security teams face alert fatigue due to overwhelming volumes of security events.
- Manual forensic investigation is slow, repetitive, and resource-intensive.
- Analysts need automated systems that can reconstruct attacks intelligently.
- Faster, evidence-based reconstruction improves response speed and decision-making.

Proposed Solution

- Introduce **CCRE** for automated cyber incident reconstruction.
- Ingests raw security logs and normalizes them into structured events.
- Reconstructs complete attack chains using strict AI-driven causal reasoning.
- Builds interactive attack graphs with critical path identification.

Objectives

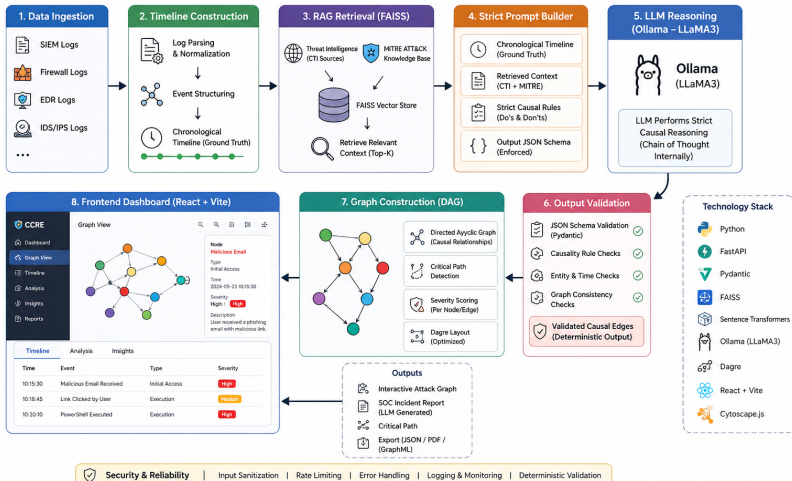
- Automate cyber incident reconstruction from raw security logs.
- Infer causal relationships using strict evidence-based AI reasoning.
- Visualize attack progression through interactive graph structures.
- Reduce investigation time and improve incident response efficiency.

Existing System vs Proposed System

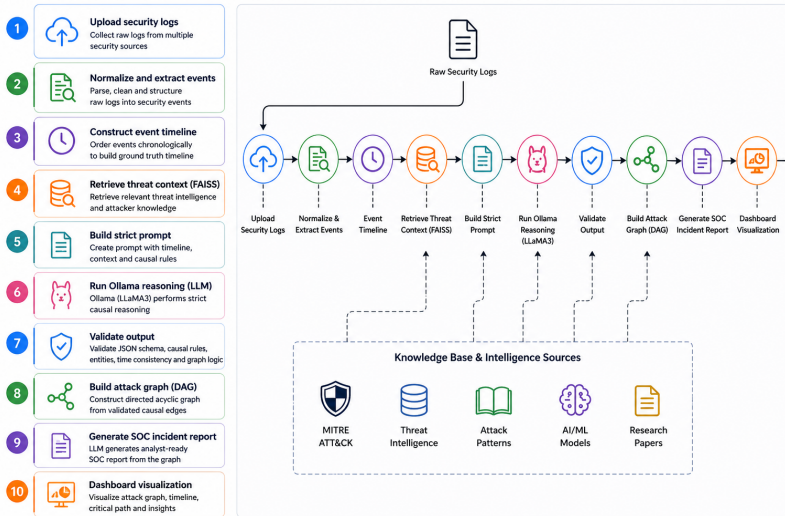
Existing System	Proposed System
Manual SIEM alert correlation	Automated incident reconstruction
Disconnected alert investigation	Connected causal attack chains
Static rule-based analysis	AI-driven causal reasoning
Analyst-dependent workflow	Deterministic evidence-based pipeline
No structured attack narrative	Grounded SOC incident report

System Architecture

CCRE – System Architecture



Methodology / Workflow



Technology Stack

Frontend

- React + Vite
- Cytoscape.js
- Dagre

Backend

- FastAPI
- Python
- Pydantic

Retrieval / Storage

- FAISS
- Neo4j

AI / ML

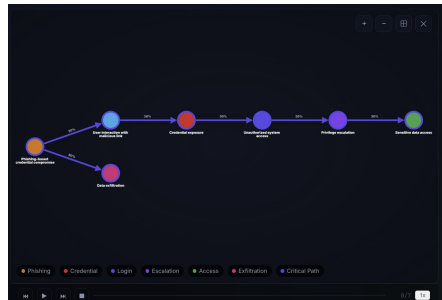
- Sentence Transformers
- Ollama (qwen2.5)

Causal Inference Engine

- Chronological timeline is treated as the ground truth.
- FAISS retrieves relevant threat context from knowledge base.
- Strict prompt combines timeline, context, and causal rules.
- Ollama performs evidence-based causal reasoning.
- Backend validates JSON structure, entities, and causal links.

Graph Model

- Nodes represent atomic security events.
- Edges represent validated causal relationships.
- Graph is structured as a directed acyclic graph (DAG).
- Critical path highlights the primary attack chain.
- Noise events are excluded for clarity and focus.



RAG + LLM Explanation

RAG Layer

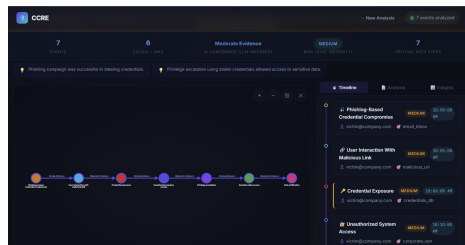
- Retrieves relevant threat context using FAISS.
- Injects MITRE-aligned attack intelligence.
- Grounds reasoning in known threat patterns.

LLM Layer

- Ollama performs strict causal reasoning.
- Generates grounded SOC incident narrative.
- Explains attack flow with evidence-based analysis.

Frontend Dashboard

- Split-pane interactive analyst dashboard
- GraphView with DAG-based attack visualization
- Timeline, Analysis, and Insights tabs
- Severity badges and critical path highlights
- Interactive node inspection and safe rendering



Use Case Scenario

- User receives a phishing email containing a malicious link.
- Credentials are stolen through fake authentication.
- Attacker performs unauthorized account access.
- Privilege escalation enables deeper system compromise.
- Sensitive data is accessed and prepared for exfiltration.
- CCRE reconstructs the complete attack chain end-to-end.

Benchmark Evaluation Results

Sample ID	Attack Scenario	Path Accuracy	Precision@K	Recall@K	Faithfulness	Latency (s)
Log 8	Ransomware (RDP Entry)	0.92	0.80	1.00	0.95	18.2
Log 9	SQL Injection	0.88	1.00	0.80	0.90	22.5
Log 10	Insider Threat (USB)	1.00	0.60	1.00	1.00	12.1
Log 11	Brute Force (O365)	0.85	0.80	0.80	0.88	25.4
Log 12	Malware (C2 Beacon)	0.90	1.00	0.90	0.92	19.8
Log 13	Cloud (SSRF/IAM)	0.78	0.80	0.60	0.85	30.2
Log 14	Supply Chain Attack	0.82	1.00	0.80	0.90	28.5
Log 15	Persistence (Task)	0.95	0.60	1.00	0.98	15.6
Log 16	Spear Phishing (Doc)	0.94	0.80	0.90	0.95	17.1
Log 17	DDoS (Resource)	0.80	0.60	0.80	0.80	21.3

Evaluation Metrics

- Precision@k for RAG retrieval relevance
- Edge Precision / Recall / F1 Score
- Path Accuracy for attack chain ordering
- Faithfulness of generated reasoning
- Groundedness against source timeline
- End-to-end inference latency

Advantages

- Faster and more reliable incident reconstruction
- Reduces analyst workload and alert fatigue
- Improves attack visibility and root cause tracing
- Generates grounded, analyst-ready security reports
- Supports faster and better incident response decisions

Limitations

- Ollama inference introduces noticeable processing latency
- Large incident timelines may exceed LLM context limits
- LLM may infer incorrect reasoning despite valid structure
- Current in-memory state limits multi-user scalability
- Concurrent requests may affect analysis consistency

Challenges Faced

- Handling noisy and incomplete security logs
- Enforcing strict JSON output from the LLM
- Preventing hallucinated entities and causal links
- Maintaining graph connectivity and visual clarity
- Balancing inference quality with response latency

Future Enhancements

- Move inference to Celery + Redis background workers
- Stream analysis using WebSockets for real-time updates
- Improve reasoning with few-shot prompt engineering
- Enforce safer structured output with constrained JSON
- Add metadata-aware FAISS retrieval for better context

Applications

- Security Operations Centers (SOCs)
- Incident response and threat hunting
- Digital forensics and breach investigation
- Enterprise cyber defense and monitoring
- Security research and attack analysis

Conclusion

- CCRE automates cyber incident reconstruction using AI-driven reasoning.
- It combines RAG, graph intelligence, and strict validation for reliable analysis.
- The system improves investigation speed, clarity, and analyst decision-making.
- CCRE provides a strong foundation for next-generation cyber forensics.

Thank You

Questions?